

Bridging Information Silos: The Role of Graph Databases in NASA's Lessons Learned and Skills Discovery Systems

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Abstract

As a sprawling organization with decades of mission data, NASA faces a significant challenge in managing "tribal knowledge" and technical expertise. This research piece explores NASA's adoption of graph database paradigms to build Knowledge Graphs (KGs). By moving away from traditional relational databases, NASA has successfully integrated disparate data silos to accelerate decision-making for future missions.

1. Introduction: The Problem of Fragmented Knowledge

NASA's primary challenge was not a lack of data, but the "siloization" of that data. Decades of engineering reports, project post-mortems (Lessons Learned), and personnel records were stored in disconnected relational databases.

Comparison and Selection Rationale:

- **Information Retrieval:** Engineers spent an estimated 20-30% of their time searching for information.
- **The "Retirement Risk":** As veteran engineers retired, their "tribal knowledge", the informal understanding of how complex systems interact, was lost.
- **Redundancy:** Without a unified view, different teams often unknowingly repeated the same technical mistakes or "reinvented the wheel."

2. Why Graph Databases? (Paradigm Analysis)

NASA evaluated relational databases (RDBMS) against the Graph paradigm. While RDBMS are excellent for structured, tabular data, they struggle with the complex, many-to-many relationships inherent in aerospace engineering.

Comparison and Selection Rationale:

- **Relationship Complexity:** In a graph database, relationships (Edges) are first-class citizens. NASA needed to map how a "Part" relates to a "Mission," which relates to an "Engineer," who possesses a "Skill." Doing this in RDBMS would require dozens of expensive "JOIN" operations, slowing performance.
- **Schema Flexibility:** Aerospace data is evolving. Graph databases are schema-optional, allowing NASA to add new types of data (e.g., telemetry or new material types) without migrating an entire database.

- **Inferencing:** Graph algorithms (like PageRank or Centrality) allow NASA to find "hidden" experts who might not have a specific job title but are central nodes in technical documentation networks.

3. High-Level Data Model and Architecture

NASA utilized Knowledge Graphs to create a "connected" view of their ecosystem. The architecture is built on a Labeled Property Graph (LPG) model.

Data Entities (Nodes):

- Lessons Learned: Specific insights gained from past missions.
- People: Engineers, scientists, and contractors.
- Topics/Metadata: Keywords, technical disciplines, and mission phases (e.g., "Launch," "Entry, Descent, and Landing").

Relationships (Edges):

- ENGINEER --[AUTHORED]--> LESSON
- LESSON --[RELEVANT_TO]--> MISSION_PHASE
- ENGINEER --[HAS_SKILL]--> PYTHON/FLUID_DYNAMICS

The Technological Stack:

1. Neo4j: Used primarily for the "Lessons Learned" Knowledge Graph. It allows engineers to query the database using Cypher, a language that can find connections such as: "Find all lessons learned regarding 'O-rings' authored by engineers who worked on the Apollo missions."
2. Memgraph: Utilized for real-time people-knowledge graphs. Because Memgraph is an in-memory database, it provides the high-speed performance required for real-time collaborative tools and identifying expertise on the fly.
3. NLP Pipeline: NASA uses Natural Language Processing (NLP) to scan unstructured PDF reports, extract entities, and automatically inject them as nodes and edges into the graph.

4. Real-World Application: Finding Hidden Skills

By applying graph algorithms to their data, NASA went beyond simple search. They used Link Prediction and Community Detection to identify skill gaps. If a project required an expert in "cryogenic propulsion," the graph could identify an engineer who hadn't officially listed that skill but was frequently cited in cryogenic research papers or linked to related mission nodes.

5. Conclusion

NASA's transition to graph databases represents a shift from "storing data" to "connecting knowledge." By leveraging Neo4j and Memgraph, NASA has reduced the time required to access critical safety information and ensured that the expertise of the past is available to the engineers of the future. This architecture serves as a

blueprint for any large-scale organization dealing with complex, interconnected information.

Sources:

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